

Agarkov Matvey



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ABOUT ME

I'm a Software Developer with over 3 years of experience, passionate about building impactful systems. I value strong foundations in computer science — data structures, algorithms, OS, networks, and design patterns — and believe communication is just as essential. My main interests include high-load distributed systems, microservices, consensus algorithms, and AI methods.

KEY COMPETENCIES

- **C++:** 20 standard, STL, threading, metaprogramming, boost
- **UI:** Qt6, QML
- **Build Systems:** Make, CMake,
- **Package System:** Conan, vcpkg
- **Testing:** Boost.Test, GoogleTest, Catch, pytest
- **Tooling:** GDB, Valgrind, perf, gprof, clang-tidy/format
- **RDBMS:** PostgreSQL, Clickhouse
- **Project Management:** Jira, Confluence
- **DevOps:** GitLab CI, GitHub Actions, Docker, Kubernetes, Jenkins
- **Distributed systems:** Apache Kafka, ZeroMQ, Redis
- **SQL:** joins, window, analytical functions
- **VCS:** Git
- **Cloud:** Yandex Cloud
- **Other Languages:** Rust, JS

PROFESSIONAL EXPERIENCE

Middle C++ Developer

Magnit Tech

Point of sale platform

Moscow, Russia, April 2025 - Present, 4 mos

- Work in Magnit Tech with the SCO (self-checkout) team, which is part of the POS development team, in collaboration with QA engineers and business analysts.
- Enhancement of cashless payment functionality (SBP, card payments), which involved extending the cashier state machine with new states and led to a 10% acceleration of the payment process on the device side.
- Improvement of the data format conversion module, resulting in the implementation of JSON-to-XML conversion and support for three new operation types.
- Took part in improving existing automated test coverage for each service and implemented integration tests with required mocks/stubs using Python and pytest.
- Optimized the Kafka-based data transmission module (15M messages/day) by introducing additional topics for existing operations, reducing average request queue time by 1.5%.

C++ Developer

T1

CAE Desktop app

Moscow, Russia, August 2022 - April 2025, 2 yr 9 mos

- Worked at T1 Integration on a CAE (Computer-Aided Engineering) desktop application integrated directly into CAD (Computer-Aided Design) systems.
- Participated in the development of middle-end logic to connect a QML-based GUI with the application core using Qt6 and C++20.
- Developed an extension to the std::ranges library (ranges_ext) that implements functionality introduced in the C++23 standard related to ranges and views.
- Participated in the development of a remote computation library using Boost.Asio, C++20 coroutines, and ZeroMQ for asynchronous communication.
- Participated in the development of UI components using Qt6 and QML, and integrated them with the middle-end layer.

EDUCATION

Bauman Moscow State Technical University, Russia

Bachelor's Degree in Electronics and Nanoelectronics

Sep 2020 - Jun 2024

GPA: 4,57/5

Bauman Moscow State Technical University, Russia

Master's Degree in Software Engineering

Sep 2024 - Jun 2026